

Excel Fundamentals

Course Map

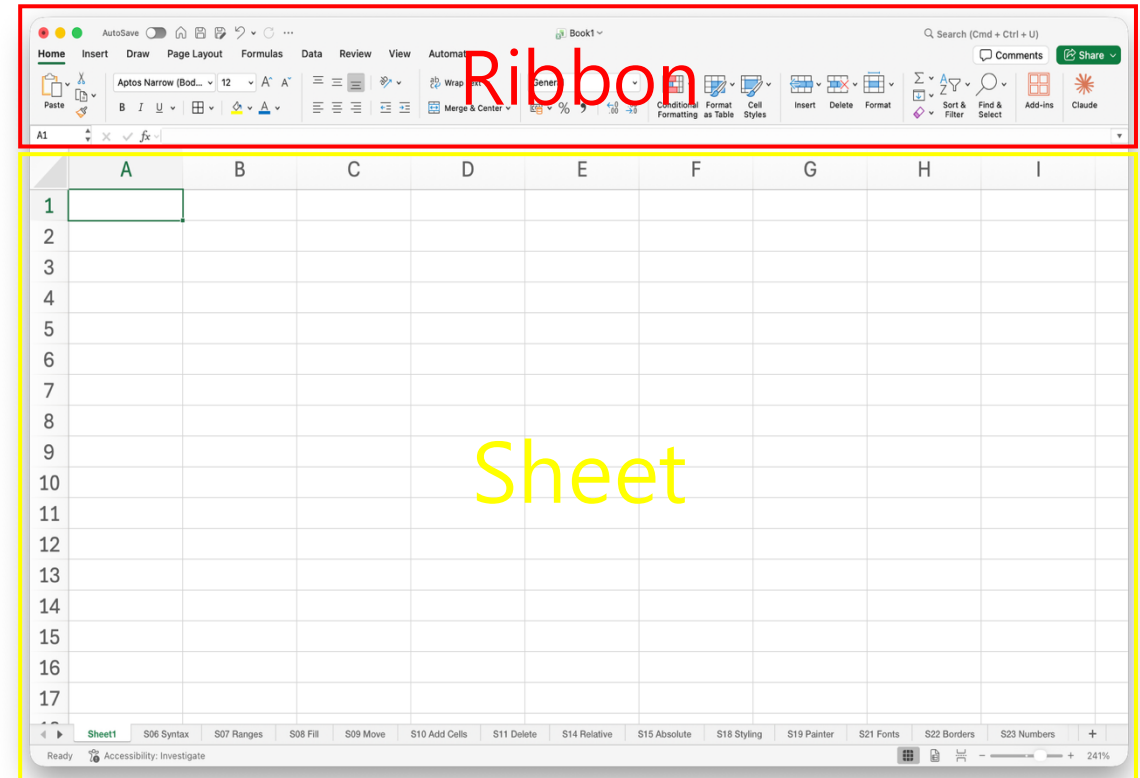
- Getting started — Introduction, Overview
- Working in the sheet — Syntax, Ranges, Fill, Move / Add / Delete cells, Undo & Redo
- Calculations — Formulas, Relative references, Absolute references
- Operators & functions — Addition, Parentheses, Functions
- Formatting — Format Painter, Colors, Fonts, Borders, Number formats, Grids, Regional settings

Introduction — What Is Excel?

- A spreadsheet program from Microsoft that organises data in columns and rows and performs mathematical functions
- Runs on Windows, macOS, Android and iOS; first released in 1985 with the core functionality largely unchanged
- Used for analysis, data entry and management, accounting, budgeting, charts, programming and financial modelling
- Why Excel: the most popular spreadsheet in the world, easy to start, a high skill ceiling, huge community support, reusable templates

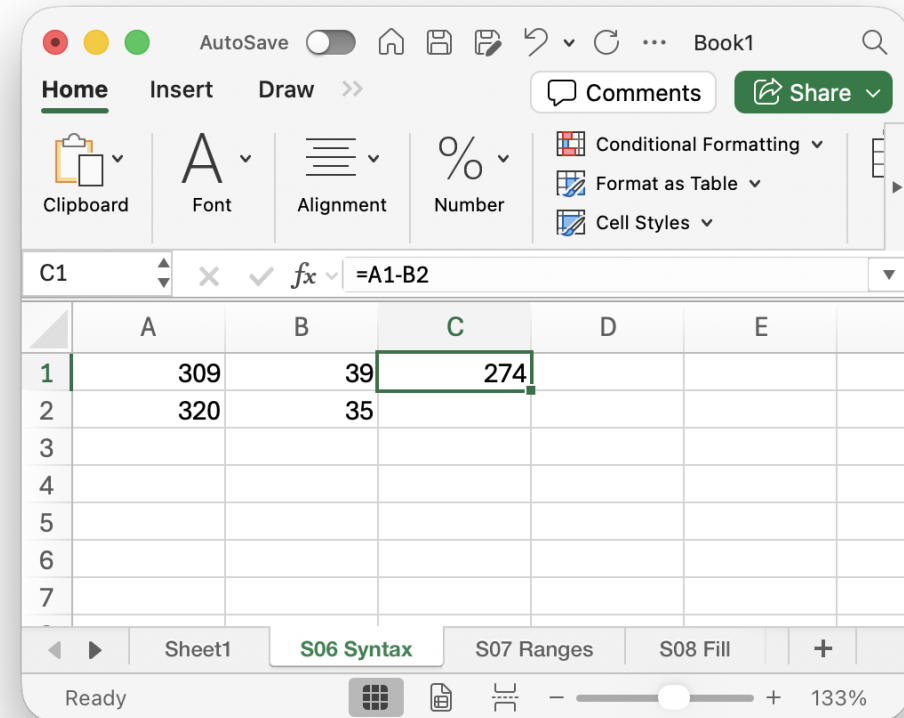
Overview — The Ribbon and the Sheet

- Excel has two main parts: the Ribbon (commands) and the Sheet (data)
- Ribbon = App launcher (nine dots, Office 365 navigation) · Tabs (menus of grouped functionality) · Groups (related commands split by a thin vertical line) · Commands (the buttons that perform actions)
- Sheet = rows and columns whose intersections form cells; cells hold numbers or text
- Every cell has a unique reference: column letter + row number, e.g. C4 — shown in the Name Box
- A workbook can hold many sheets: Shift + F11 adds one, right-click a tab to rename (e.g. Raw Data, Data Structure, Data Visualization)



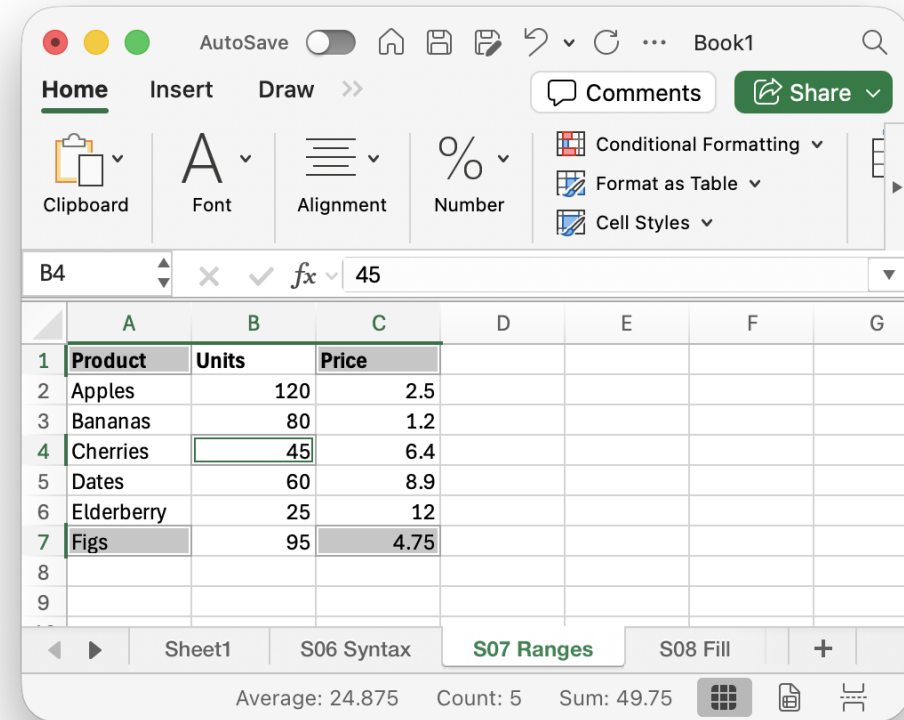
Syntax — How a Formula Is Written

- Every formula starts with the equal sign = typed into the selected cell, followed by the calculation
- Step by step: select a cell → type = → select a cell or type a value → type an operator → select another cell or value → press Enter
- =1+1 returns 2
- Cell values are described as reference(value), for example A1(309)
- Example: with A1(309), A2(320), B1(39), B2(35), the formula C1 = A1-B2 returns 274



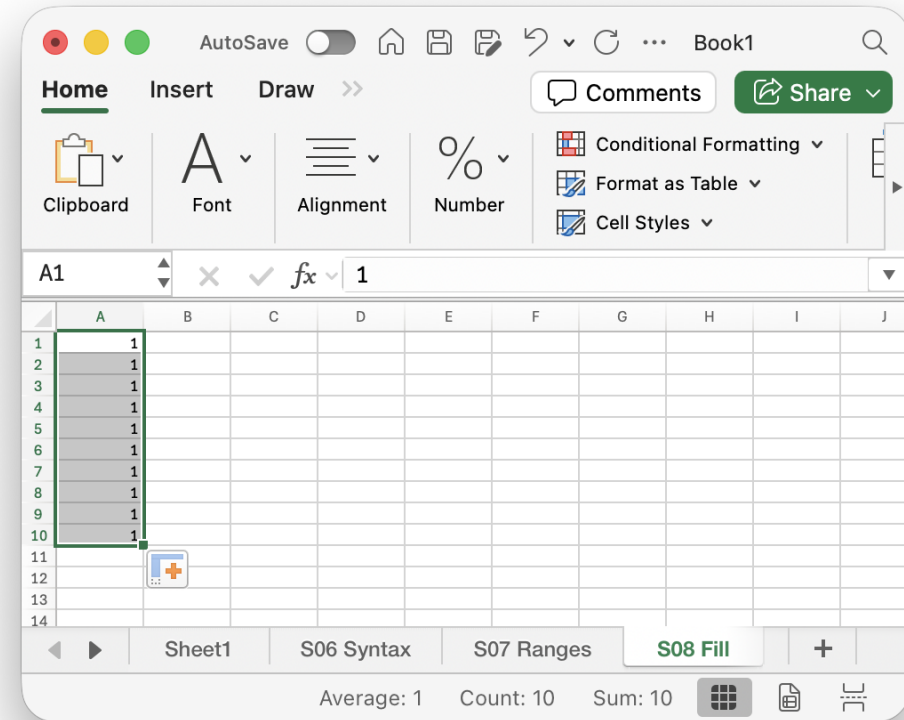
Ranges — Selecting Cells

- A range is a selection of cells — the basis for almost every Excel operation
- Four selection operations: a single cell, multiple cells, a whole column, a whole row
- Multiple cells: hold CTRL (Command on Mac) and left-click each cell
- A column is selected by clicking its letter; a row by clicking its number
- The Name Box shows the reference of the current selection — and you can type a reference into it to jump there



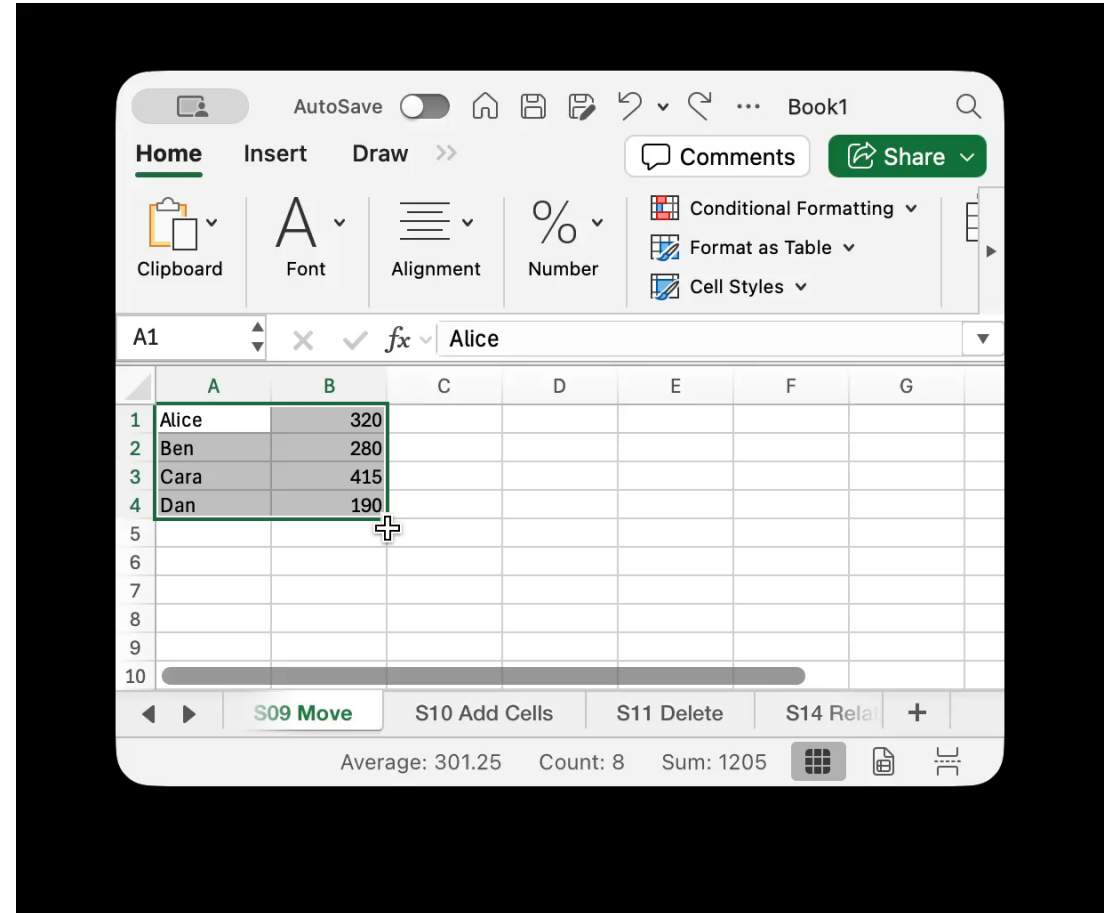
Fill — Filling Ranges Automatically

- Filling populates a range with values so you do not have to type each entry
- Used for copying, sequences, dates and functions
- The fill handle is the small square in the bottom-right corner of a cell; the pointer becomes a thin cross
- Click and hold the handle, then drag over the range — for example from A1 down to A1:A10
- Filling copies both numbers and text: A1(1) filled to A1:A10 gives ten copies of 1



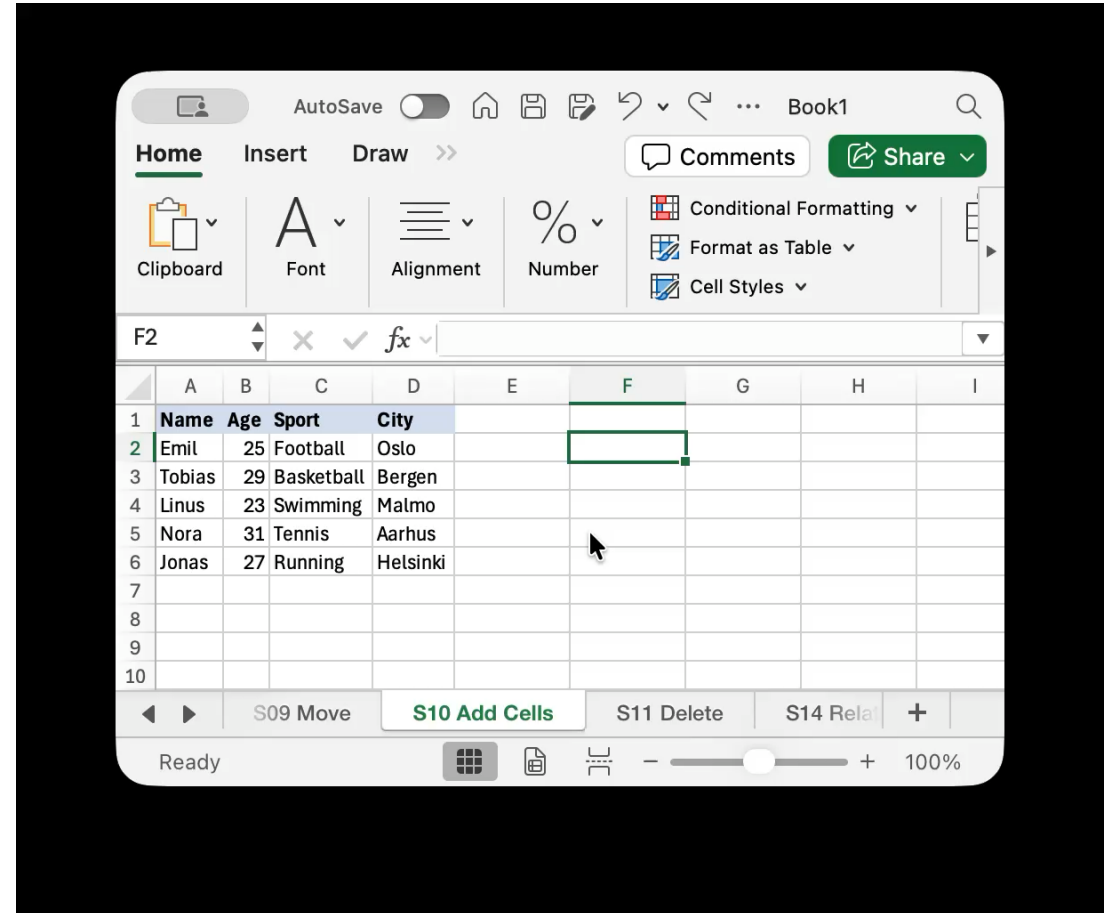
Move Cells — Drag & Drop, Cut & Paste

- Cells can be moved in two ways: drag and drop, or cut and paste
- Drag and drop: mark the range (e.g. A1:B4), hover the border until the move cursor appears, then drag it to the new position (e.g. B2:C5)
- Cut and paste: mark the range, press CTRL + X, select the destination cell, press CTRL + V
- Moving data creates room for headers — always give the data context with labels so the sheet is easy to read



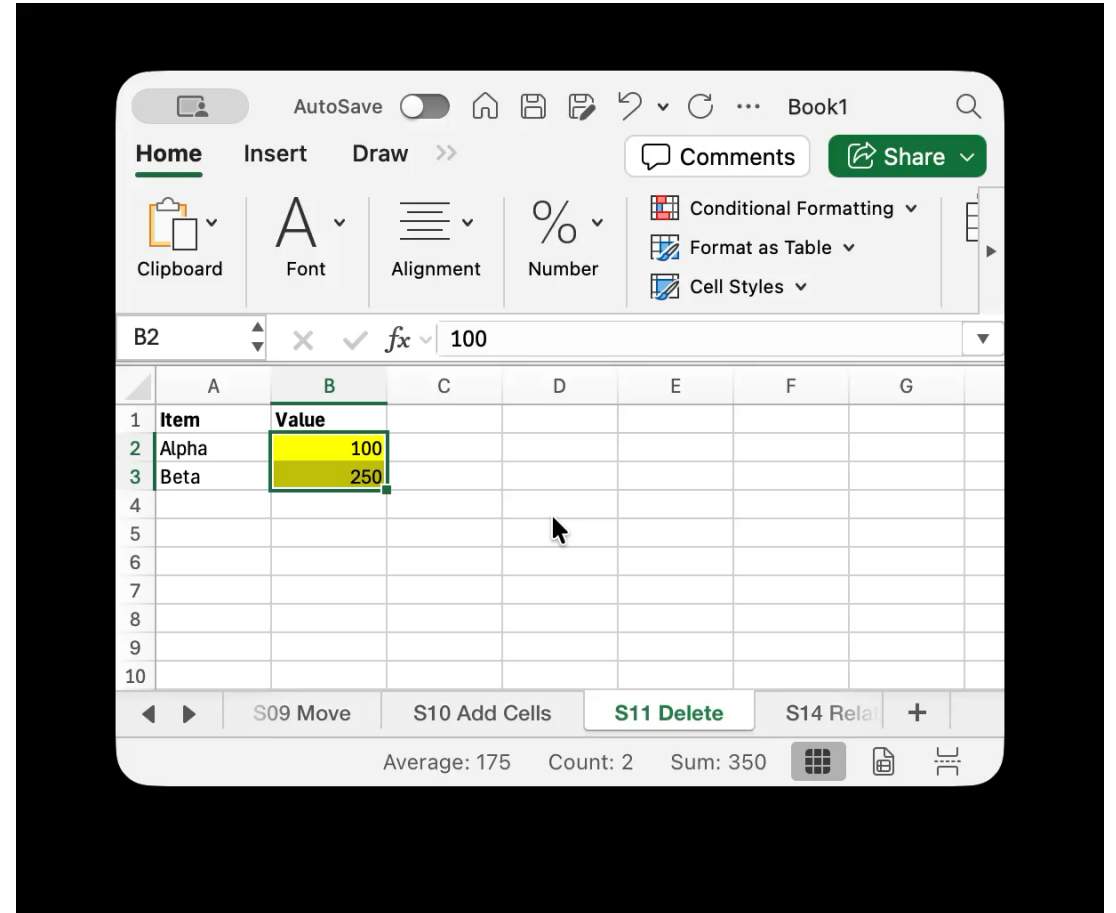
Add Cells — Insert Columns and Rows

- Right-click a column letter and choose Insert Columns — the new column appears where you clicked
- Right-click a row number and choose Insert Rows — the new row appears where you clicked
- Existing data shifts right or down automatically, so nothing is lost
- Use this to add a field you forgot (a new Trainer column) or a missing record (a new data row)



Delete Cells — Values vs. Formatting

- Select the cells and press the Delete key to remove their contents
- Delete removes the value only — the formatting stays in place
- A cell with a colour fill keeps that colour after its value is deleted
- Clear the formatting separately when you want the styling removed as well



Undo and Redo

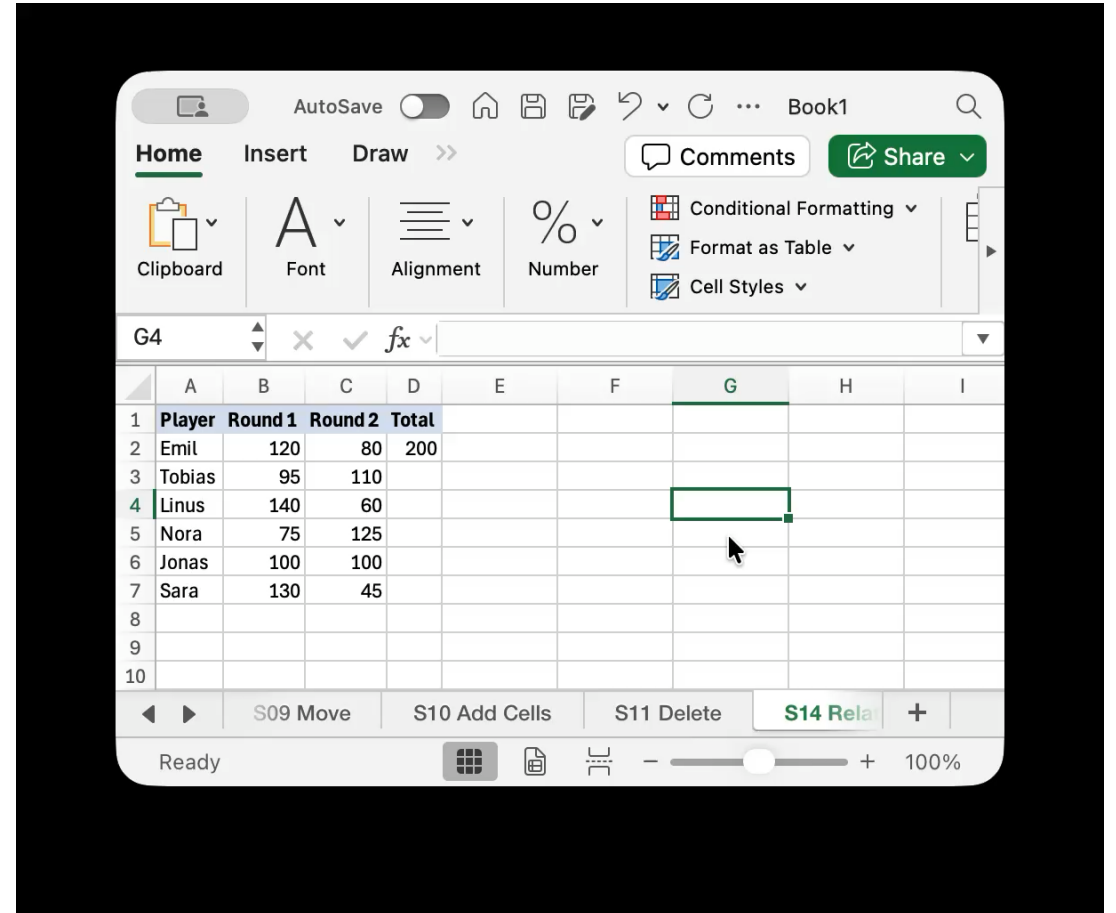
- Undo reverses your last action — the Ribbon button, or CTRL + Z (Command + Z on Mac)
- Redo reverses an Undo — the Ribbon button, or CTRL + Y (Command + Y); available only after an Undo
- Works for actions inside the sheet: deleting a formula, adding a column, removing a row
- Does not work for File-menu actions such as deleting a sheet, saving, or changing options
- Learn the keyboard shortcuts — they save a lot of time

Formulas — Calculating with Cells

- Formulas do the maths and always begin with =, for example =1+1, =2*2, =4/2
- Cells can be the inputs: select C1, type =, click A1, type +, click A2, press Enter
- With A1(2) and A2(4), C1 returns 6; change + to * and the result becomes 8
- Swap the operator for -, * or / to change the calculation
- Calculating with references rather than typed numbers is the core Excel skill — results update when the inputs change

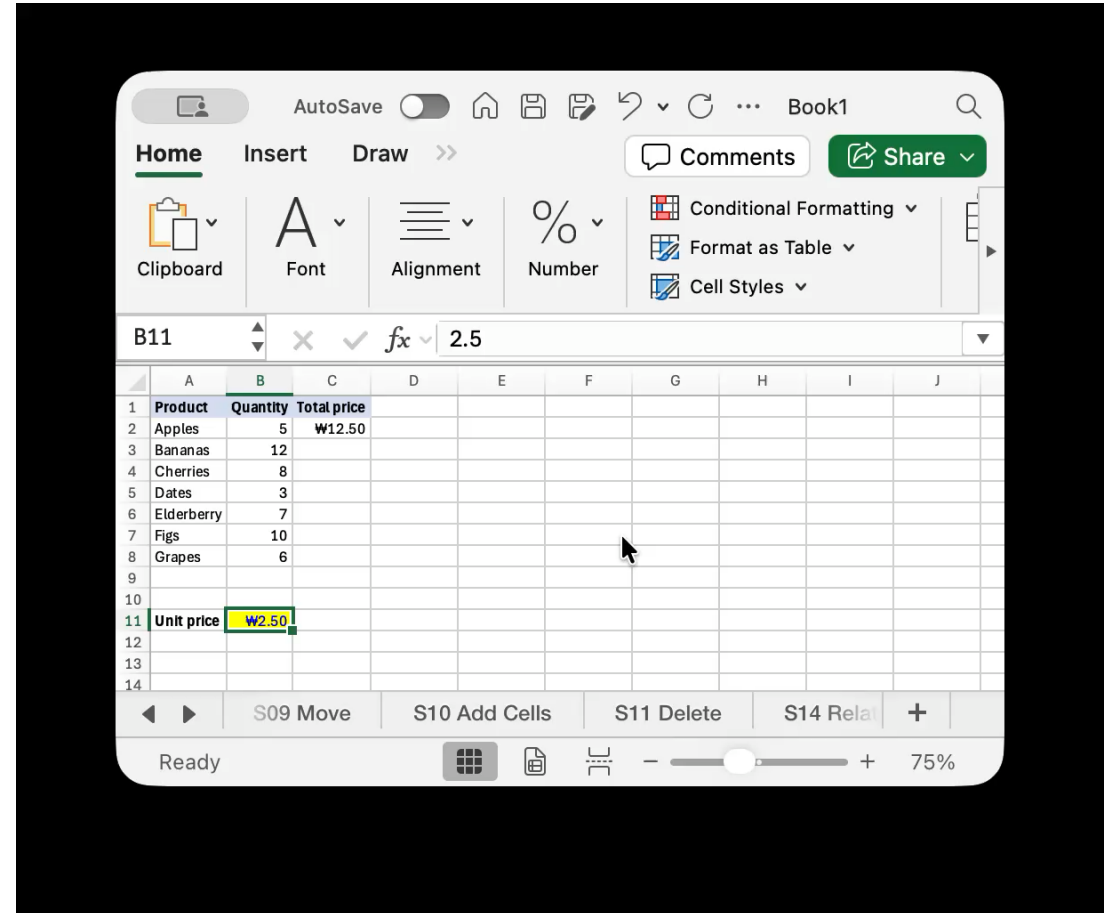
Relative References

- Excel has two reference types: relative and absolute
- References are relative by default and written without a dollar sign, e.g. A1
- A relative reference shifts as the fill function copies the formula down or across
- Example: D2 =B2+C2 filled to D7 becomes D3 =B3+C3 ... D7 =B7+C7
- This is what lets a single formula serve an entire column



Absolute References

- An absolute reference uses the dollar sign \$ to lock a reference so filling cannot change it
- \$A\$1 — column and row locked · \$A1 — column locked, row relative · A\$1 — row locked, column relative
- Example: with the unit price in B11, C2 =B2*\$B\$11 filled down keeps pointing at B11
- Use absolute references for constants such as prices, tax rates and conversion factors



Addition Operator

- Addition uses the + symbol, or the SUM function
- Steps: type = → select a cell → type + → select another cell → press Enter
- Manual entries: =5+5 returns 10
- Two cells: =A1+A2 · Several cells: =A1+A2+A3+A4+A5
- SUM is faster for longer ranges: =SUM(A1:A5) gives the same result

Parentheses — Order of Operations

- Parentheses () change the order of a calculation — whatever is inside is calculated first
- $=10+5*2$ returns 20, because $5*2$ is calculated first and added to 10
- $=(10+5)*2$ returns 30, because (15) is calculated first and then multiplied by 2
- A formula can hold several groups: $=(10+5)+(2*4)+(4/2)$
- Cells work inside parentheses too, for example $=(A1+A2)*B5$

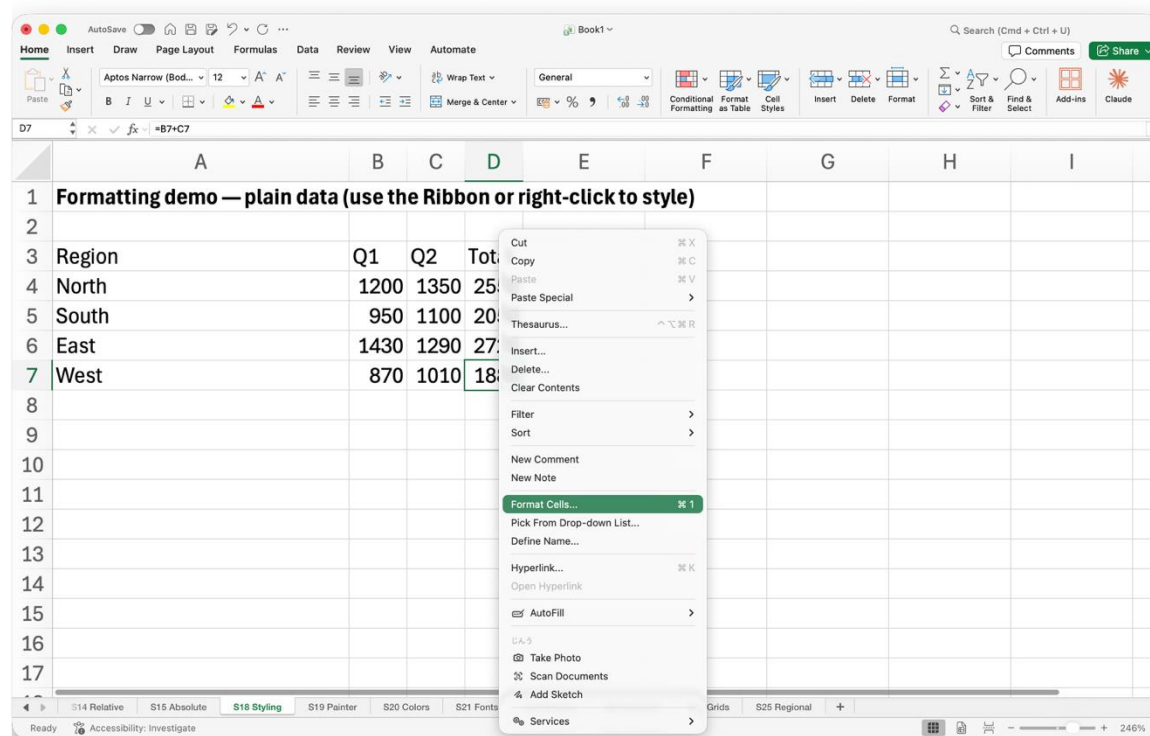
Functions — Premade Formulas

A function is typed as = plus its name, applied to a range in parentheses — for example =SUM(A1:A5)

Function	What it does	Function	What it does
=SUM	Adds the values in a range	=IF	Returns one value if TRUE, another if FALSE
=AVERAGE	Calculates the arithmetic mean	=COUNT	Counts cells containing numbers
=MIN / =MAX	Returns the smallest / largest value	=COUNTA	Counts all non-empty cells
=CONCAT	Joins the content of several cells	=SUMIF / =COUNTIF	Adds or counts based on a condition

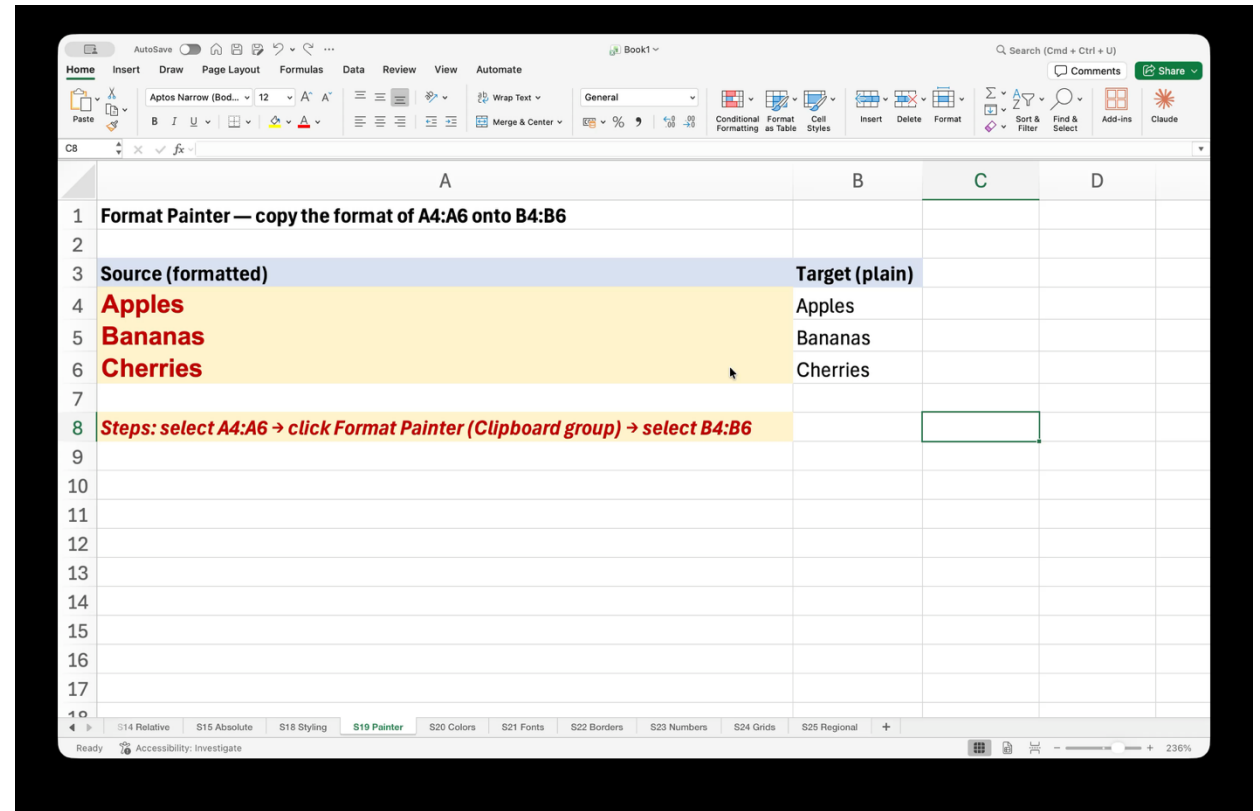
Formatting — Why and Where

- Formatting changes how cells look: colours, fonts, font sizes, borders, number formats and grids
- Why format: it makes a spreadsheet easier to read and understand, and more presentable
- The most used styling functions are Colors, Fonts, Borders, Number formats and Grids
- Two ways to reach the styling commands: the Ribbon (expand it with the caret icon on the right for more commands), or right-click a cell for the formatting menu



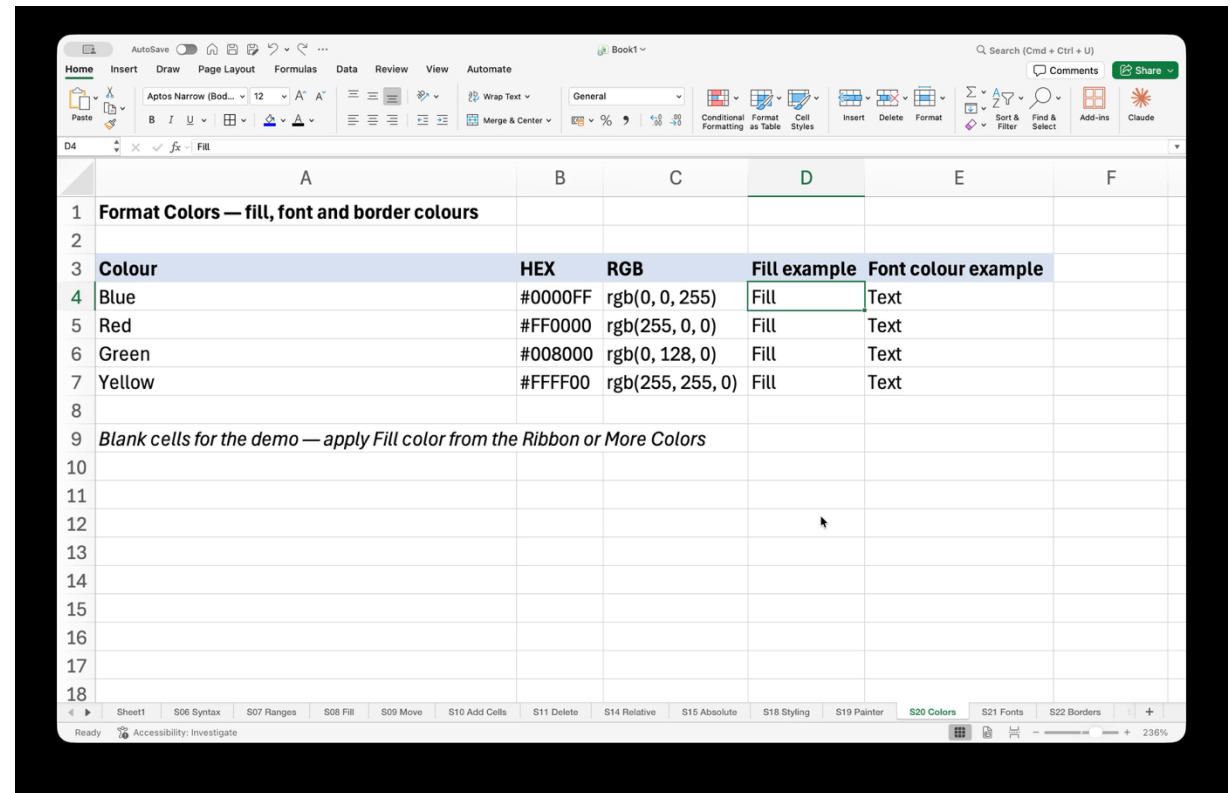
Format Painter — Copy Formatting

- The Format Painter copies formatting from one cell to another — a big time-saver
- Found in the Ribbon, in the Clipboard group
- How to use it: select the source cell, click Format Painter, then select the target cell or range
- A dotted border around the source cell shows the format is copied and ready to paste
- Works for single cells and for ranges, e.g. copying font characteristics from A4:A6 to B4:B6



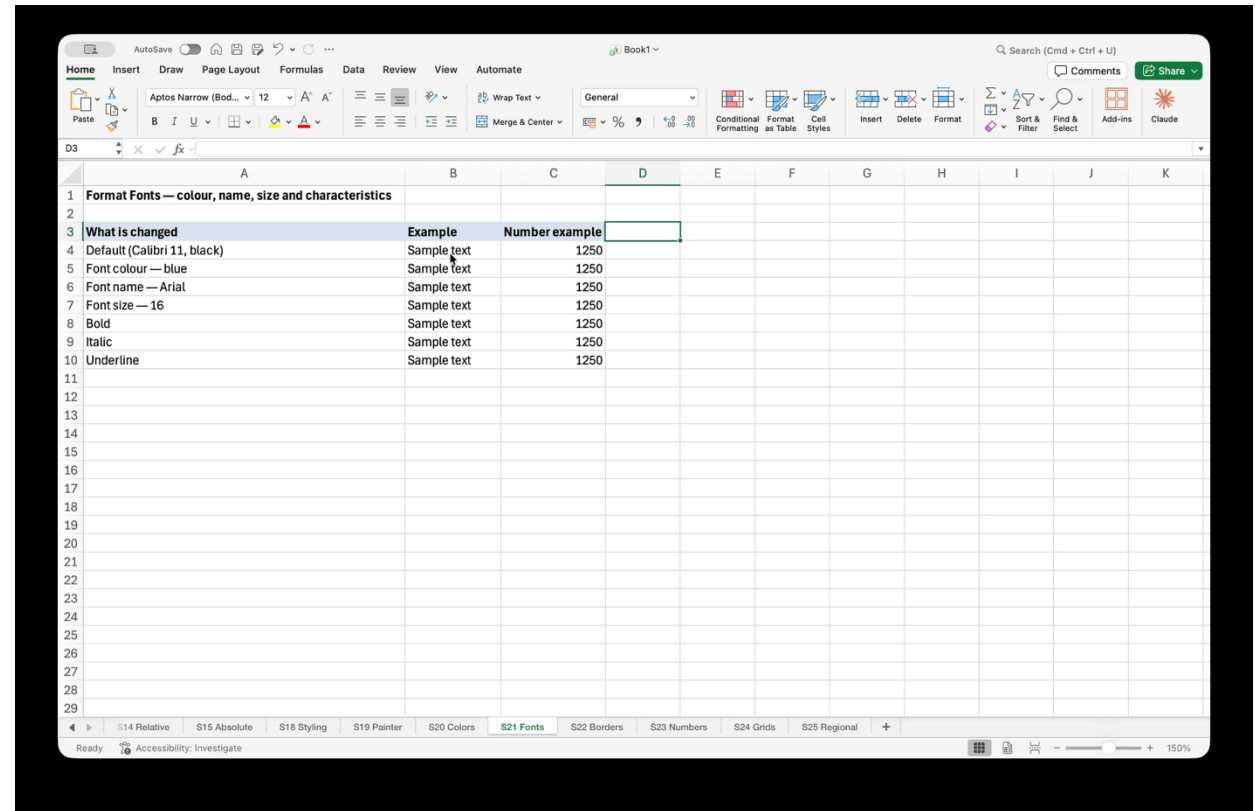
Format Colors

- Colours can be picked from theme and standard colours, or entered as HEX or RGB codes via More Colors
- Hexadecimal: #RRGGBB, each pair from 00 to FF — #0000FF is blue
- RGB: rgb(red, green, blue), each value from 0 to 255 — rgb(0,0,255) is blue
- Colours apply to cells (Fill color), text (Font color) and borders
- The Fill color button remembers the colour you used last time



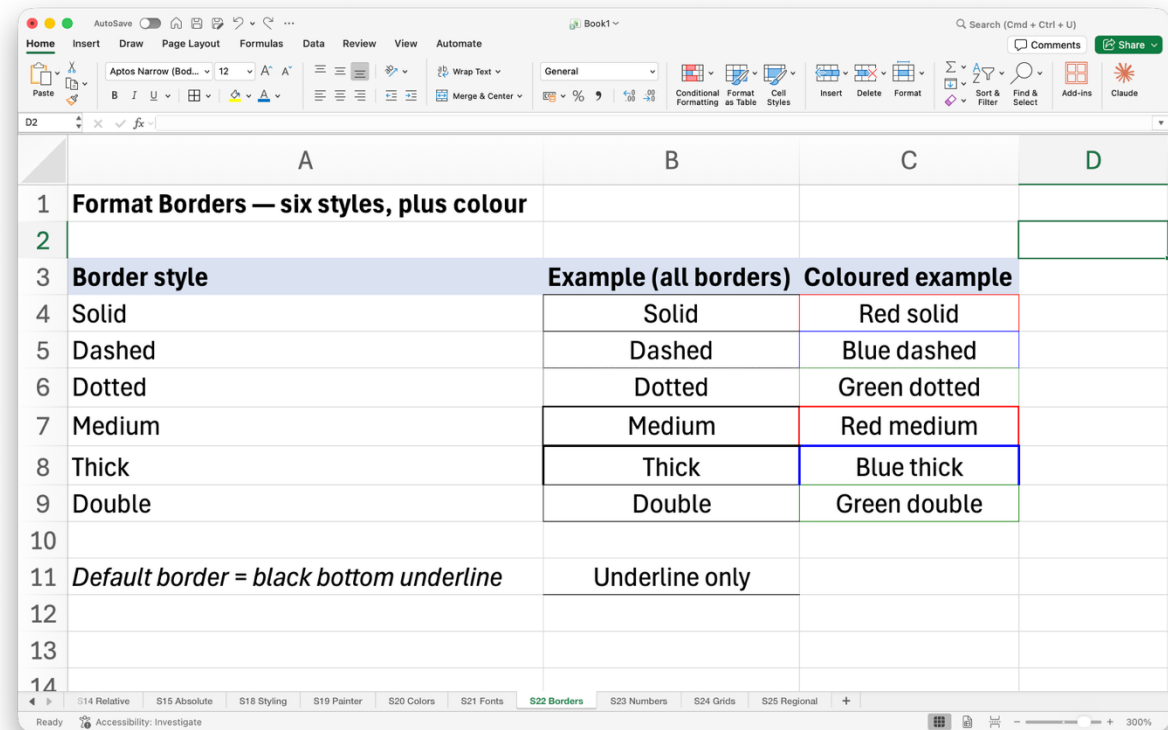
Format Fonts

- Fonts can be formatted in four ways: colour, font name, size and characteristics
- Font colour: the default is black; select the cell, pick a Font color, then type — it applies to numbers and text alike
- Font name: the Excel default is Calibri; select a range and pick another font, e.g. Arial
- Font size and characteristics — bold, italic and underline — are set from the Font group in the Ribbon
- Both the Font color and Font name commands remember your last choice



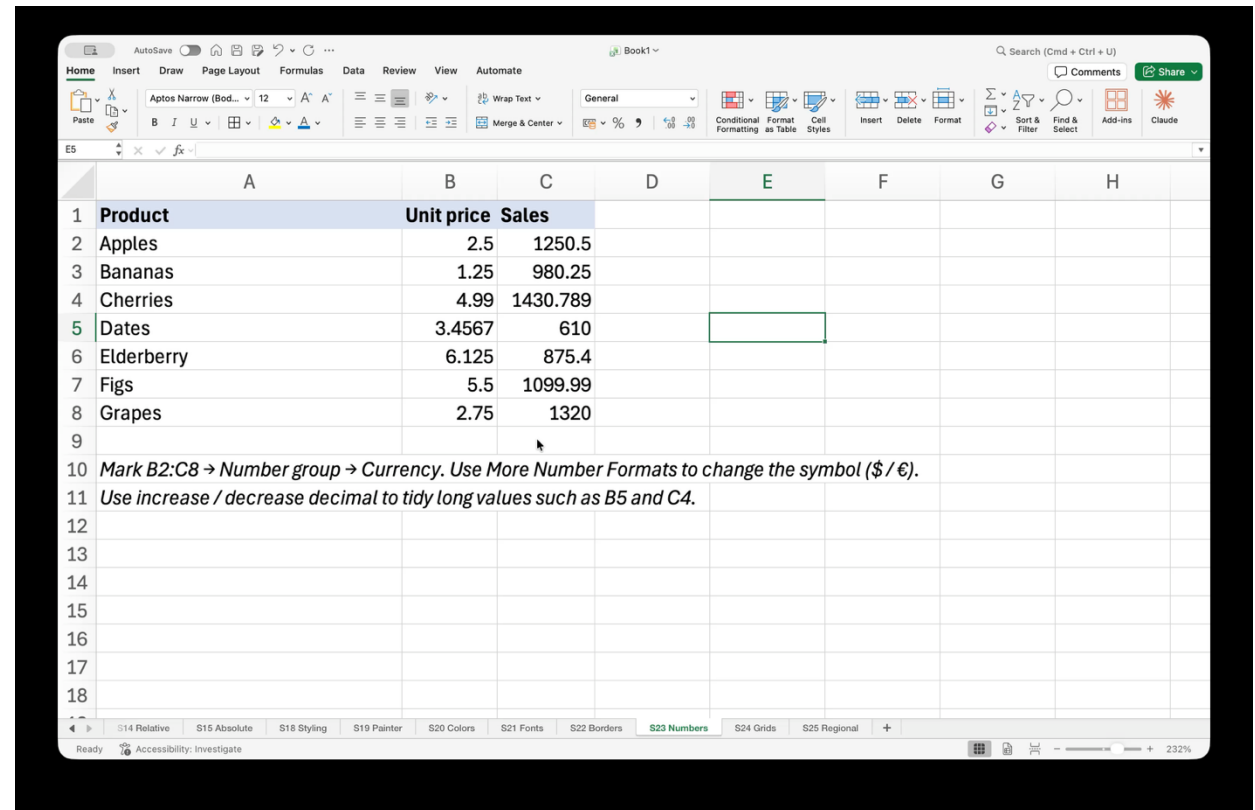
Format Borders

- Borders can be added, removed, coloured and restyled — useful for readability and for emphasising key points
- The Borders menu sits in the Ribbon, in the Font group; the button remembers your last border
- The default border is a black underline; the option arrow next to the button shows all border types
- Border colour: choose a colour in the Border Color menu before applying the border
- Six border styles are available: solid, dashed, dotted, medium, thick and double



Number Formats

- The default number format is General; change it to make data explainable and to prepare it for functions
- Common formats: General, Number, Currency, Time — set from the Number format dropdown in the Number group
- Example: mark the range B2:C8 and choose Currency; Excel uses your local currency by default
- More Number Formats at the bottom of the menu lets you switch the currency symbol, e.g. \$ or EUR
- Use the increase/decrease decimal buttons to tidy up long decimal values



Format Grids

- Gridlines are displayed by default, but they can be hidden
- Why remove them: a cleaner, more readable and more presentable sheet
- How to: click View in the Ribbon, then uncheck Gridlines
- Removing gridlines affects the whole spreadsheet, not just the selected cells

